杭州电子科技大学学生考试卷（ A ）卷

考试课程	通信电路与系统	考试日期	2022 年 6 月 22 日	成 绩			
课程号	A040095s	教师号		任课教师姓名	陈瑶、林弥、李航、刘国华、刘艳、张伟、孙朋飞、苏江涛、关阳阳		
考生姓名		学号（8 位）		年级		专业	

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题
得分						

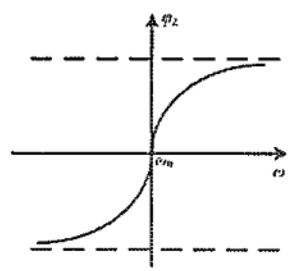
I. Single Choice (3' each)

1. Which of the following circuits is not part of AM receiving system? (D)

A. Local Oscillator B. Detector

C. High Frequency Amplifier D. Modulator

2. A LC resonant circuit has impedance characteristic curve shown as below. When operating at the resonant frequency ω_0, the resonant impedance of the circuit shows (C).



A. Capacitive B. Inductive

C Minimum Resistor D. Maximum Resistor

3. Consider that the power consumption of power amplifiers is the main energy consumption of wireless communication system. In order to improve efficiency and realize green communication, which kind of power amplifier is the best choice? (C)

A. Class-A B. Class-B

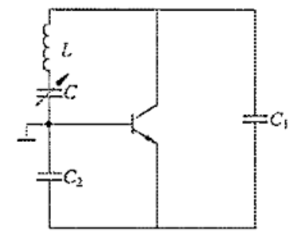
C. Class-C D. Class-AB

4. If f_c and f_L are the frequencies of input carrier signal and local oscillation signal respectively, then the output intermediate frequency signal is f_I ($=f_L-f_c$) through a mixer. The frequency component $2f_c-f_L=f_I+300$Hz. What kind of interference does it belong to? (A)

A. Whistle Interference B. Side-channel Interference

C. Cross-modulation Interference D. Intermodulation Interference

5. A feedback LC oscillator is given as below. What kind of circuit is it? (C)



A. Colpitts Circuit B. Hartley Circuit

C. Clapp Circuit D. Seiler Circuit

6. In parallel mode crystal oscillator, the crystal acts as a/an (A).

A. Inductor B. Capacitor

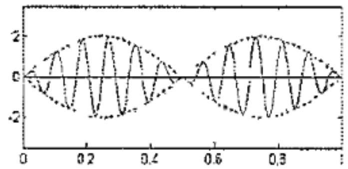
C. Resistor D. Varactor

7. Which of the following characteristics of diode or transistor cannot be used for amplitude modulation? (D).

A. Base Modulation Characteristic B. Collector Modulation Characteristic

C. Switch Characteristic D. Varactor Characteristic

8. The time domain waveform of the output modulated signal is shown as below. Which kind of modulation is it? (B).



A. Standard AM　　　　B. DSB AM

C. SSB AM　　　　D. FM

9. An angle modulated wave is $u(t)=10\cos(2\pi\times10^6 t+5\cos2\pi\times10^3 t)$ (v). What is the maximum frequency shift? (　C　)

A. 10 KHz　　　　B. 10 MHz

C. 5 KHz　　　　D. 5 MHz

10. For angle modulated wave, which of the following description is wrong? (　B　)

A. For standard FM wave, the maximum frequency shift will change only with the amplitude of modulation signal

B. For standard PM wave, the maximum frequency shift will change only with the amplitude of modulation signal

C. For standard FM wave, the maximum phase shift will change both with the amplitude and frequency of modulation signal

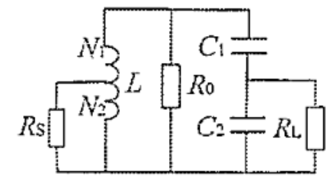
D. For standard PM wave, the maximum phase shift will change only with the amplitude of modulation signal

II. Calculation

1. (10') The LC parallel resonant circuit is shown as below, the source resistor R_S and load R_L are partially accessed by auto-transformer and capacitor divider. Known that L=1μH, Q_0=100, C_1=C_2=10pF, R_S=10kΩ, R_L=10kΩ. Suppose that R_L and R_S achieve impedance matching.

(1) Calculate the resonance frequency f_0 and the resonant resistance R_0 (R_L and R_S are negligible).

(2) Calculate N_2/N_1, loaded quality factor Q_L and bandwidth $B_{0.7}$.



Solution:

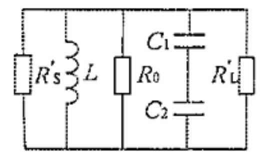
(1) $f_0=\dfrac{1}{2\pi\sqrt{LC_\Sigma}}$　　$C_\Sigma=\dfrac{C_1C_2}{C_1+C_2}=5pF$

$f_0=\dfrac{1}{2\pi\sqrt{10^{-6}\times5\times10^{-12}}}Hz\approx71.2MHz$ ……… ……… …(2')

$R_0=Q_0\omega_0L=100\times2\pi\times71.2\times10^6\times10^{-6}\Omega=44.7K\Omega$ …………(2')

(2) $n_L=\dfrac{C_1}{C_1+C_2}=0.5$　　$R_L'=\dfrac{R_L}{n_L^2}=40K\Omega$

$R_\Sigma=R_0\parallel R_L'=\dfrac{R_0R_L'}{R_0+R_L'}=21.1K\Omega$



$n_S=\dfrac{N_2}{N_1}$　　$R_S'=\dfrac{R_S}{n_S^2}$　　$R_S'=R_\Sigma$　　$n_S=\sqrt{\dfrac{R_S}{R_\Sigma}}=\sqrt{\dfrac{10}{21.1}}=0.69$ ....(2')

$Q_L=\dfrac{R_S'\|R_\Sigma}{\omega_0L}=\dfrac{0.5\times21.1\times10^3}{2\pi\times71.2\times10^6\times10^{-6}}=23.6$ ....(2')

$B_{0.7}=\dfrac{f_0}{Q_L}=\dfrac{71.2}{23.6}MHz=3MHz$ ....(2')

Simply calculation:

$R_L'=\dfrac{R_L}{n_L^2}=40K\Omega$　　$R_S'=R_L'$　　$n_S=\sqrt{\dfrac{R_S}{R_L'}}=\sqrt{\dfrac{10}{40}}=0.5$

$R_0\parallel R_L'\parallel R_S'=\dfrac{44.7\times20}{44.7+20}=13.8K\Omega$

$Q_L=\dfrac{R_0\parallel R_L'\parallel R_S'}{\omega_0L}=\dfrac{13.8\times10^3}{2\pi\times71.2\times10^6\times10^{-6}}=30.9$　　$B_{0.7}=\dfrac{f_0}{Q_L}=\dfrac{71.2}{30.9}MHz=2.3MHz$

2. (11') A Class-C resonant power amplifier with operating at critical point state. Assuming the output power P_o=3W, E_C=13V, the saturation voltage of transistor is U_{ces}=1V, the current conduction angle θ=60°, $\alpha_0(60°)$=0.218, $\alpha_1(60°)$=0.391.

(1) Please calculate the DC power P_{DC}, power dissipation at the collector P_c, collector efficiency η_c.

(2) In order to achieve the climate goal of carbon neutrality and emission peak, what measures can we take to improve the efficiency η_c of class-C power amplifier for RF energy saving? What do you think about the influences of index of high power for a power amplifier on wireless communication system, natural environment, and human health?

Solution:

(1)

$$u_{c1m} = E_C - U_{ces} = (13-1)\text{V} = 12\text{ V}$$

$$R_C = \frac{1}{2}\frac{u_{c1m}^2}{P_o} = \frac{12^2}{2\times3}\Omega = 24\Omega$$

$$I_{c1m} = \frac{u_{c1m}}{R_C} = \frac{12}{24}A = 500mA$$

$$I_{cmax} = \frac{I_{c1m}}{\alpha_1(\theta)} = \frac{500}{0.391}mA = 1.28A$$

$$I_{c0} = I_{cmax}\alpha_0(\theta) = 1.28\times0.218A = 279mA$$

$$P_{DC} = E_C I_{c0} = 13\times279mW = 3.63W$$ ....(2')

$$P_c = P_{DC} - P_o = (3.63-3)W = 0.63W$$ ....(2')

$$\eta_c = \frac{P_o}{P_{DC}} = \frac{3}{3.63}\times100\% = 82.6\%$$ ....(2')

(2) Provide at least one correct way and explain the way for improving efficiency can get scores. ....(3')

$$\eta_c = \frac{1}{2}\gamma\xi = \frac{1}{2}\frac{\alpha_1(\theta)}{\alpha_0(\theta)}\frac{u_{c1m}}{E_C}$$

Way-1: appropriately increase current conduction angle θ, usually ranges from 50°~70°

Way-2: appropriately increase the collector voltage u_{c1m}, u_{cemin} decreases with the increase of u_{c1m} and this change will lead to transistor working at saturation region.

Way-3: appropriately increase the collector resistance R_C, based on the load characteristic of PA, the efficiency at critical point state isn't the maximum value. The maximum efficiency of class-C appears at the over-voltage status.

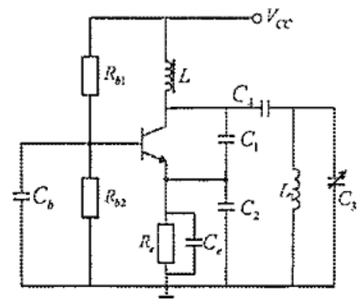
Understand about high power. .. ..(2')

At one hand, high power is helpful to realize long-distance signal transmission. At the other hand, higher power means stronger radiation power, which is harmful to environment and human body.

3. (10') The oscillation circuit is shown as below, assuming V_{CC}=12V, R_{b1}=10kΩ, R_{b2}=5kΩ, R_e=1kΩ, C_b=0.2μF, C_1=330pF, C_2=510pF, C_3=(2~30)pF, C_4=20pF, L_1=1uH, L=200uH.

(1) Draw the high frequency AC equivalent circuit and explain the type of oscillator.

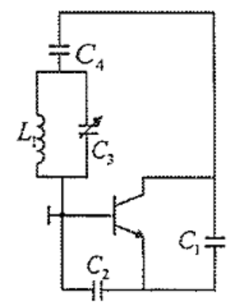
(2) Calculate the oscillating frequency ranges f_{0min}, f_{0max} and the feedback factor F.



Solution:

(1) Seiler circuit or the second improved capacitor feedback LC oscillator. ....(2')

Draw correct equivalent circuit. ....(2')



(2)

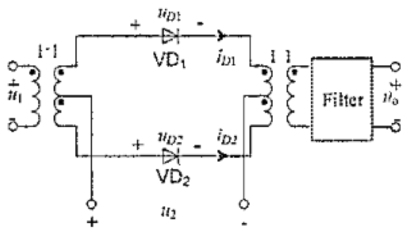
$$C_\Sigma = C_3 + \frac{1}{\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_4}} \approx C_3 + C_4 = 22\sim50pF$$ ....(2')

$$f_0 = \frac{1}{2\pi\sqrt{L_1C_\Sigma}} \approx \frac{1}{2\pi\sqrt{10^{-6}\times(22\sim50)\times10^{-12}}}Hz = 22.5\sim33.9MHz$$ ....(2')

$$F \approx \frac{C_1}{C_1+C_2} = \frac{330}{330+510} = 0.39$$ .. . .(2')

$\frac{2}{\pi}$ cos

4. (20') The diode multiplier circuit is shown as below. Assume that the diodes VD1 and VD2 are ideal component and exactly the same. $u_1(t)=u_{1m}\cos\Omega t\cos\omega_c t$, $u_2(t)=u_{2m}\cos\omega_L t$, $u_{2m}>>u_{1m}$, the conductance of diodes is g_D.

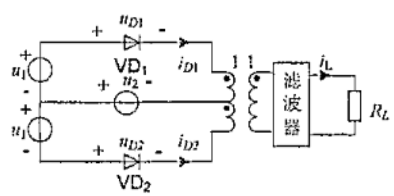


(1) If the equivalent resistance of the filter at the center frequency f_0 is R_L, write the expression of output voltage $u_o(t)$ and list the frequency component of $u_o(t)$.

(2) Please draw the frequency spectra of $u_1(t)$, $u_2(t)$, and $u_o(t)$.

(3) What type of filter should be used at u_o terminal? Please write the center frequency f_0 and the bandwidth B of the filter.

Solution:



(1)

$u_{D1}=u_1+u_2$ ....(1')

$u_{D2}=-u_1+u_2$ ....(1')

$i_{D1}=g_D K_1(\omega_L t)(u_1+u_2)$ ....(1')

$i_{D2}=g_D K_1(\omega_L t)(-u_1+u_2)$ ....(1')

$u_1=u_{1m}\cos\Omega t\cos\omega_c t=\frac{1}{2}u_{1m}[\cos(\omega_c-\Omega)t+\cos(\omega_c+\Omega)t]$ ....(1')

$K_1(\omega_L t)=\frac{1}{2}+\frac{2}{\pi}\cos\omega_L t-\frac{2}{3\pi}\cos 3\omega_L t+\frac{2}{5\pi}\cos 5\omega_L t-...$ ....(2')

$i_{D1}-i_{D2}=2g_D K_1(\omega_L t)\cdot u_1$

$=g_D u_{1m}[\cos(\omega_c-\Omega)t+\cos(\omega_c+\Omega)t](\frac{1}{2}+\frac{2}{\pi}\cos\omega_L t-\frac{2}{3\pi}\cos 3\omega_L t+\frac{2}{5\pi}\cos 5\omega_L t-...)$

..........(2')

$u_o(t)=i_L R_L=\frac{2}{\pi}g_D u_{1m}R_L[\cos(\omega_c-\Omega)t+\cos(\omega_c+\Omega)t]\cos\omega_L t$

$=\frac{1}{\pi}g_D u_{1m}R_L[\cos(\omega_c-\Omega-\omega_L)t+\cos(\omega_c-\Omega+\omega_L)t+\cos(\omega_c+\Omega-\omega_L)t+\cos(\omega_c+\Omega+\omega_L)t]$

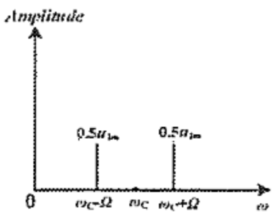
..........(2')

The frequency component of $u_o(t)$ is $\omega_c-\Omega\pm\omega_L$ and $\omega_c+\Omega\pm\omega_L$.......... .....(1')

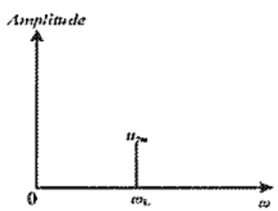
(2) Frequency spectrum of $u_1(t)$, ...(2')

Frequency spectrum of $u_2(t)$, ...(1')

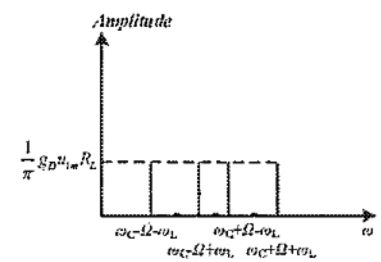
Frequency spectrum of $u_o(t)$ ....(2')



Frequency spectrum of $u_1(t)$



Frequency spectrum of $u_2(t)$



Frequency spectrum of $u_o(t)$

(3) Bandpass filter. ....(1')

$f_0=\frac{\omega_c}{2\pi}$ ....(1')

$B=\frac{2(\Omega+\omega_L)}{2\pi}=\frac{\Omega+\omega_L}{\pi}$ ....(1')

5. (14') The circuit is shown as below in fig. (a), assuming C=0.05uF, the frequency spectrum of input signal $u_i(t)$ is given in fig. (b).

(1) Please write the mathematical expression of the input signal $u_i(t)$, draw the time domain waveform and explain what kind of modulation signal is it?

(2) For fig. (a), what kind of circuit is it? What is the frequency f_o of the output signal $u_o(t)$?

(3) How to choose the load resistance R_L to prevent diagonal distortion? How to choose the load resistance R_L to acquire high detection efficiency?

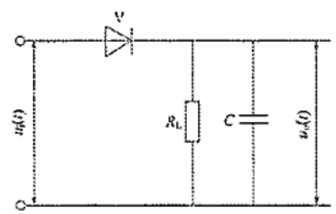


fig. (a)

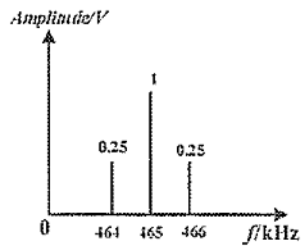


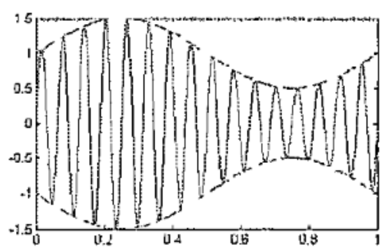
fig. (b)

Solution:

(1) $u_i(t)$ =cos2π×465×10³t+0.25cos2π×466×10³t+0.25cos2π×464×10³t (v)

=cos2π×465×10³t(1+0.5cos2π×10³t) (v)..(2')

Standard AM wave or AM wave..(2')



Time domain waveform...........................(2')

(2)

Large signal peak envelope detector..........................(2')

$$f_o = \frac{\Omega}{2\pi} = 1KHz$$ ....(2')

(3)

m_a=0.5

Ω=2π×10³ rad/s

$$R_L < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\Omega C} = \frac{\sqrt{0.75}}{0.5} \times \frac{1}{2\pi \times 10^3 \times 0.05 \times 10^{-6}} \Omega = 5.5K\Omega$$ ....(2')

ω_c=2π×465×10³ rad/s

$$R_L >> \frac{1}{\omega_c C} = \frac{1}{2\pi \times 465 \times 10^3 \times 0.05 \times 10^{-6}} \Omega = 6.8\Omega$$ ....(2')

6. (5') The phase modulated signal $u_{PM}(t)$=10cos(2π×10⁶t+10cos2π×10³t) (v).

(1) Find out the carrier wave frequency f_c, modulation signal frequency F, and PM factor m_p?

(2) Calculate the maximum frequency shift Δf_P and the effective bandwidth B of the PM signal.

Solution:

(1) $u_{PM}(t)$=10cos(2π×10⁶t+10cos2π×10³t)=U$_{cm}$cos(ωct+ $m_p$cosΩt)

$$f_c = \frac{\omega_c}{2\pi} = 1MHz$$ ....(1')

$$F = \frac{\Omega}{2\pi} = 1KHz$$ ....(1')

m_p=10...(1')

(2)

$$\Delta f_P = \frac{\Delta\omega_P}{2\pi} = \frac{m_P\Omega}{2\pi} = \frac{10 \times 2\pi \times 10^3}{2\pi} Hz = 10KHz$$ ....(1')

$$B = 2(m_p+1)F = 2(\Delta f_p + F) = 2(10+1)KHz = 22KHz$$ ....(1')

杭州电子科技大学学生考试卷（ B ）卷

考试课程	通信电路与系统	考试日期	2022 年 6 月 22 日	成 绩			
课程号	A0400955	教师号		任课教师姓名	陈瑾、林弥、李航、刘国华、刘艳、张伟、孙朋飞、苏江涛、关阳阳		
考生姓名		学号（8 位）		年级		专业	

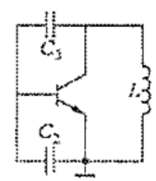
题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题	第七大题
得分							

I. Single Choice (3' each)

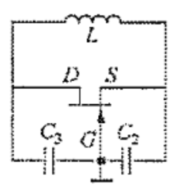
1. A serial LC circuit resonates at the frequency of f_0. When the actual frequency f is larger than f_0 ($f>f_0$), the circuit shows (A).

A. Inductive B. Capacitive C. Resistive D. Short

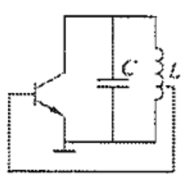
2. By using the phase equilibrium condition, determine which of the oscillation circuits shown below is possible. (D)



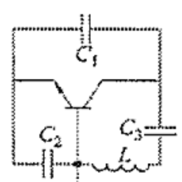
A.



B.



C.



D.

3. Which of the conduction angle (CCA) 2θ is classified as Class C power amplifier? (D)

A. $2\theta=360^0$ B. $180^0<2\theta<360^0$
C. $2\theta=180^0$ D. $2\theta<180^0$

4. What is not the reason why Quartz crystal has high frequency stability (B)

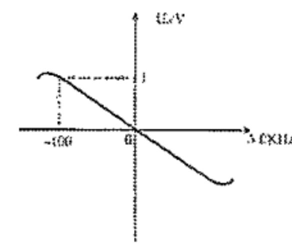
A. It has low temperature coefficient

B. It can be seen as capacitor

C. It has high Q-factor.

D. It has low access factor.

5. The frequency discrimination characteristics curve has been shown below. Known that the output voltage of the frequency discriminator is $u_o(t)=0.5\cos4\pi\times10^3t$ (V), the frequency discriminating trans-conductance g_d and maximum frequency shift Δf_m of input signal are (A)

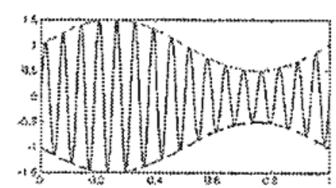


A. -1/100 (V/KHz),50 KHz B. -1/100 (V/KHz),100 KHz
C. -1/200 (V/KHz),50 KHz D. -1/200 (V/KHz),100 KHz

6. Which of the following circuits is not part of AM receiving system? (D)

A. Local Oscillator B. Detector
C. High Frequency Amplifier D. Modulator

7. For the standard AM signal shown below, the modulation index m_a is (C)



A. 1 B. 0.2

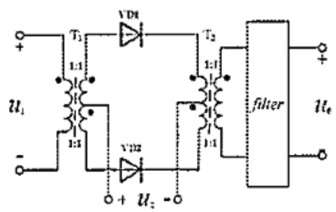
C. 0.5 D. 2

8. A broadcasting radio signal at 550 kHz is received by a receiver with a 465 kHz IF(intermediate frequency). Another signal at 1480 kHz is also received at the same time. Why does the 1480 kHz signal appear? (A)

A. Mirror image interference B. IF interference

C. Higher order side-channel interference D. Random Noise

9. The diode multiplier circuit is shown below. Assume the I/V curve characteristics of diode VD1 and VD2 are exactly the same, $u_1 = u_{1m}\cos\omega_C t$, $u_2 = u_{2m}\cos\omega_L t$, $u_{2m} >> u_{1m}$, the intermediate frequency $\omega_I = \omega_L - \omega_C$. What type of filter should be used at u_o terminal? (B)



A. Low pass filter B. Band pass filter

C. Band stop filter D. High pass filter

10. For standard FM wave, which of the following statement is correct? (B)

A. The carrier amplitude will change with the modulated signal

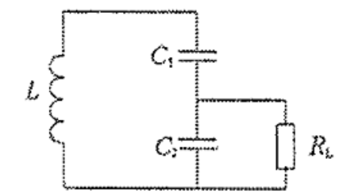
B. The carrier frequency will change with the modulated signal

C. The carrier frequency and amplitude both change with the modulated signal

D. The carrier frequency and amplitude both remain constant

II. Calculation

1. (13') The parallel LC resonant circuit is shown below with L=20uH, $C_1=C_2$=10pF, Q_0=100, R_L =20kΩ, please calculate the resonant frequency f_0, the resonant resistance R_0, quality factor load Q_L, bandwidth $B_{0.7}$.



$$C_\Sigma = \frac{C_1 \cdot C_2}{C_1 + C_2} = 5pF \quad (2')$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 15.9MHz \quad (2')$$

$$R_0 = Q_0\omega_0 L = 200K\Omega \quad (2')$$

$$n = \frac{C_1}{C_1 + C_2} = \frac{1}{2} \quad (1')$$

$$R_L' = \frac{1}{n^2}R_L = 4R_L = 80K\Omega \quad (2')$$

$$Q_L = \frac{R_\Sigma}{\omega_0 L} = \frac{R_0 // R_L'}{\omega_0 L} \approx 28.6 \quad (2')$$

$$B_{0.7} = \frac{f_0}{Q_L} = 556kHz \quad (2')$$

2. (14') A resonant power amplifier with operating at critical point state. Assuming $E_C = 24$ V, current conduction angle $\theta = 60°$, α_0 (60°) = 0.218, α_1 (60°) = 0.391, collector current pulse maximum value $I_{cmax} = 2.2$ A, collector voltage utilization factor $\xi = 0.9$.

(1) Please calculate output power P_{out}, power of power supply P_{DC}, collector efficiency η_c and collector equivalent load resistor R_c.

(2) If the equivalent load resistor R_c changing suddenly to short. How to change of output power P_{out}, power supply P_{DC}. Please analyze the reasons of changing and potential damage.

$$U_{c1m} = \xi \times E_c = 0.9 \times 24 = 21.6V \quad (1')$$

$$I_{c1m} = \alpha_1(\theta) \times I_{cmax} = 0.391 \times 2.2 \approx 0.86A \ (1')$$

$$P_{out} = \frac{1}{2} I_{c1m} U_{c1m} = \frac{1}{2} \times 21.6 \times 0.86 \approx 9.29W \quad (2')$$

$$I_{c0} = \alpha_0(\theta) \times I_{cmax} = 0.218 \times 2.2 = 0.48A \ (1')$$

$$P_{DC} = E_c \times I_{c0} = 24 \times 0.48 \approx 11.52W \quad (1')$$

$$\eta_c = \frac{P_{out}}{P_{DC}} = \frac{9.29}{11.52} \approx 80.6\% \ (2')$$

$$R_c = \frac{U_{c1m}}{I_{c1m}} = \frac{21.6}{0.86} \approx 25.1\Omega \quad (1')$$

其它条件不变时，集电极等效负载电阻 Rc 短路，功放工作到欠压状态，输出功率也会降低至 0；(3')

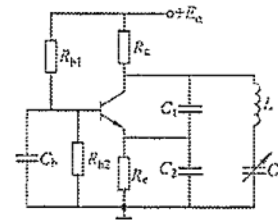
原因是输出电压基波分量会迅速减小到 0，虽然集电极电流基波分量会缓慢增加，但二者乘积仍然为 0，故输出功率为 0。所有的电源功率都用来发热，严重时会烧毁功放管。(2')

3. (11') The AC equivalent circuit of Clapp oscillator is shown below. Please answer the following questions:

(1) Assuming the value of L=100 μH, C_1=C_2=2000 pF, C=100-120 pF, calculate the resonant frequency f_0.

(2) Calculate the feedback coefficient.

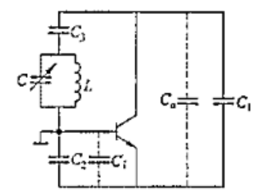
(3) To raise the frequency stability, based on Clapp oscillator, the Seiler oscillator can be obtained. Please draw the AC equivalent circuit of the Seiler oscillator circuit. And write the advantages of Seiler oscillator relative to Clapp oscillator.



(1) $f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{LC}} \times \frac{1}{2\pi \times \sqrt{100\times10^{-6}\times(100\sim120)\times10^{-12}}} = (1.45\sim1.59)\times10^6$ Hz (3')

(2) F=U_f/U_o=C_1/(C_2+C_1)=1/2 (2')

(3)



(3')

（可不画 C_i C_o）

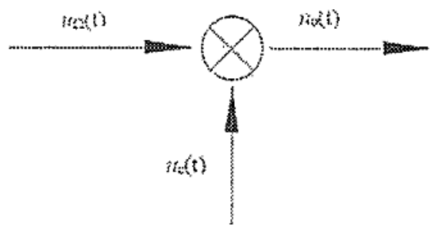
(1) Output amplitude is relatively stable when the oscillation frequency is tuned;

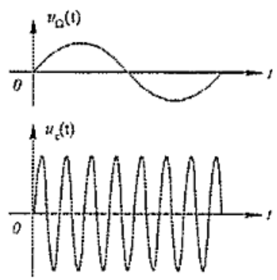
(2) Tuning range is relatively wide, the coverage coefficient of frequency is usually high, generally about 1.6 to 1.8. In practice, it has been widely used in broadband system. (3')

4. (11') The realization of u_o(t) is shown below. $u_\Omega(t) = U_{\Omega m}\cos(\Omega t)$ and $u_c(t) = U_{cm}\cos(\omega_c t)$ are modulation signal and high-frequency carrier, respectively. Please answer the following questions:

(1) If $u_i(t)$=cos100πt×cos1000πt (v), please draw the waves of u_Ω(t), u_c(t) and u_o(t), respectively. Please draw the frequency spectrum of u_o(t).

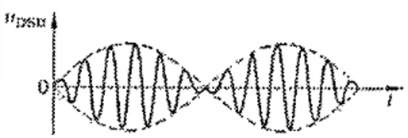
(2) Calculate the total power consumed on the unit resistance.



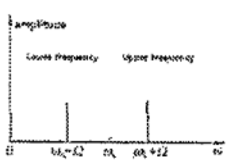


(各 1',共 2'

具体图形可自行判断)



(3')



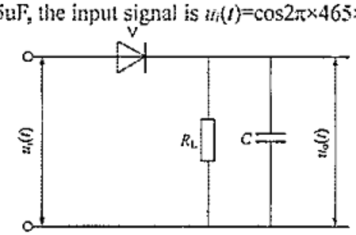
(3')

（注意横纵坐标）

(2) $u_i(t)$=cos100πt×cos1000πt=1/2×(cos900πt+ cos1100πt)

$$P=\frac{1}{2}\times(\frac{1}{2})^2+\frac{1}{2}\times(\frac{1}{2})^2=\frac{1}{4}\text{w} \qquad (3)$$

5. (12') The circuit of large signal peak envelope detector is shown as below, assuming C=0.05uF, the input signal is $u_i(t)$=cos2π×465×10^3t+0.5cos2π×10^{3}t cos2π×465×10^3t (v).



(1) Please briefly describe the working principle of the circuit.

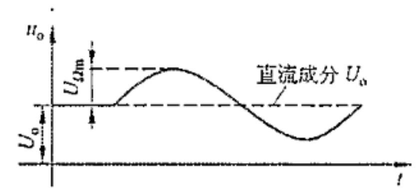
(2) Calculate the modulation factor m_a. After filtering, draw the output signal waveform.

(3) How to choose the load resistance R_L to prevent diagonal distortion?

(4) How to choose the load resistance R_L to acquire high detection efficiency?

(1)二极管工作在开关状态，正向导通电阻很小，当 $u_i>u_o$ 时，电容 C 充电很快。电阻 R_L 很大，当 $u_i<u_o$ 时，电容放电较慢。利用快充慢放的原理使输出信号跟随输入信号包络的变化，并最终得到调制信号，完成检波。 (3' 画图解释亦可)

(2) $u_i(t)$=cos2π×465×10^3t(1+0.5cos2π×10^{3}t) m_a=0.5 (2')



(2')

(3)

$\Omega=2\pi\times10^3$ rad/s

$$R_L<\frac{\sqrt{1-m_a^2}}{m_a}\times\frac{1}{\Omega C}=\frac{0.75}{0.5}\times\frac{1}{2\pi\times10^3\times0.05\times10^{-6}}\Omega=4.77K\Omega$$ (3')

(4) $\omega_C=2\pi\times465\times10^3$ rad/s

$$R_L>>\frac{1}{\omega_C C}=\frac{1}{2\pi\times465\times10^3\times0.05\times10^{-6}}\Omega=21.5\Omega$$ (2')

6. (9') Known that the modulation frequency $u_\Omega(t)$=2.5cos$4\pi\times10^3 t$ (v), the carrier signal $u_c(t)$=10cos$2\pi\times10^6 t$ (v), and maximum frequency shift Δf_{max}=100 kHz, please:

(1) Calculate the maximum phase shift m_f.

(2) Write down the mathematical expression of the output FM signal $u_{FM}(t)$.

(3) Calculate the effective bandwidth B_{FM}

(1) $m_f=\Delta f_{max}/f=100\times10^3\text{Hz}/2000\text{ Hz}=50$ (3')

(2) $u_{FM}(t)=U_{cm}\cos(\omega_C t+m_f\sin\Omega t)=10\cos(2\pi\times10^6 t+50\sin4000\pi t)$ (3')

(3) $B_{FM}=2(m_f+1)f=2\times(50+1)\times2000\,Hz=204\,kHz$ (3')